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CH.SC.U4CSE25276

Array

Q.N0:1

Problem Description:

How many Y's did a Roman Centurion make a day in cold hard Lira? About a C's worth! Turns out, Martians gave Rome the idea for their number system. Use the conversion charts below to help translate some Martian numbers!

Note, that unlike the Roman Numerals, Martian Numerals reuse symbols to mean different values. B can either mean '1' or '100' depending on where it appears in the number sequence.

Input Format:

You will receive a list of numbers in a data file, one number per line, up to 5 lines at a time (with a minimum of 1 line). No number will exceed 1000, or be less than 1.

Output Format:

Print the output in a separate lines contains convert the numbers from Arabic (1,2,3...10...500...1000) to Martian (B,BB,BBB...Z...G...R) numerals.

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
26	161
67	627

186 408 500	816 401 501
EXPECTED OUTPUT	EXPECTED OUTPUT
ZZWB PZWBB BPZZZWB BGWBBB G	BPZB GBZZWBB GBBBZWBB BGB GB

Mandatory Test Cases

Test Case 1	Test Case 2	Test Case 3
KEYWORD	KEYWORD	KEYWORD
char buf[]	buf[i++]='R';	while(n>=10)

Code:

```
#include <stdio.h>

void convert(int n)
{
    char buf[100];
    int i = 0;
```

```
while(n >= 1000)
{
    buf[i++] = 'R';
    n = n - 1000;
}
```

```
while(n >= 500)
{
    buf[i++] = 'G';
    n = n - 500;
}
```

```
while(n >= 400)
{
    buf[i++] = 'B';
    buf[i++] = 'G';
    n = n - 400;
}
```

```
while(n >= 100)
{
    buf[i++] = 'B';
    n = n - 100;
}
```

```
while(n >= 90)
{
    buf[i++] = 'Z';
    buf[i++] = 'B';
    n = n - 90;
}
```

```
while(n >= 50)
{
    buf[i++] = 'P';
    n = n - 50;
}
```

```
while(n >= 40)
{
    buf[i++] = 'Z';
    buf[i++] = 'P';
    n = n - 40;
}
```

```
while(n >= 10)
{
    buf[i++] = 'Z';
    n = n - 10;
}
```

```
while(n >= 9)
{
    buf[i++] = 'B';
}
```

```
    buf[i++] = 'K';
    n = n - 9;
}

while(n >= 5)
{
    buf[i++] = 'W';
    n = n - 5;
}

while(n >= 4)
{
    buf[i++] = 'B';
    buf[i++] = 'W';
    n = n - 4;
}

while(n >= 1)
{
    buf[i++] = 'B';
    n = n - 1;
}

buf[i] = '\0';

printf("%s\n", buf);
}

int main()
{
    int n[5];
    int count = 5;

    printf("How many input values:\n");
    printf("%d\n", count);

    printf("Input value:\n");

    for(int i = 0; i < count; i++)
    {
        scanf("%d", &n[i]);
    }

    printf("\nOutput:\n");

    for(int i = 0; i < count; i++)
    {
        convert(n[i]);
    }

    return 0;
}
```

Output:

Test_case1:

How many input values:

Input value:

26

67

186

408

500

Output:

ZZWB

PZWBB

BPZZZWB

BWBBB

G

Test_case 2:

Input value:

161

627

816

401

501

Output:

BPZB

GBZZWBB

GBBBZWB

BB

GB